

WASP-52b

R-0415

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_181115 · created 2026-07-19T07:03:57+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458437.59353 +/- 0.00029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2164 +/- 0.0011
Transit depth (Rp/Rs)^2	4.68 +/- 0.046 [%]
Orbital Inclination (inc)	85.11 +/- 0.24
Airmass coefficient 1 (a1)	0.56 +/- 0.035
Airmass coefficient 2 (a2)	0.499 +/- 0.092
Scatter in the residuals of the lightcurve fit is	6.15 %
Best Comparison Star	#2 - [383, 300]
Optimal Method	PSF photometry
Transit Duration (day)	0.06023 +/- 0.00077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.2164 ± 0.0011 vs 0.1646 (deviation 7.38σ)
Duration fit vs published	0.06023 d vs 0.0758 d (deviation 3.18σ)
Mid-time O-C	-10.8 min
Depth SNR	30.8
Residual scatter	6.15 %

Light curve

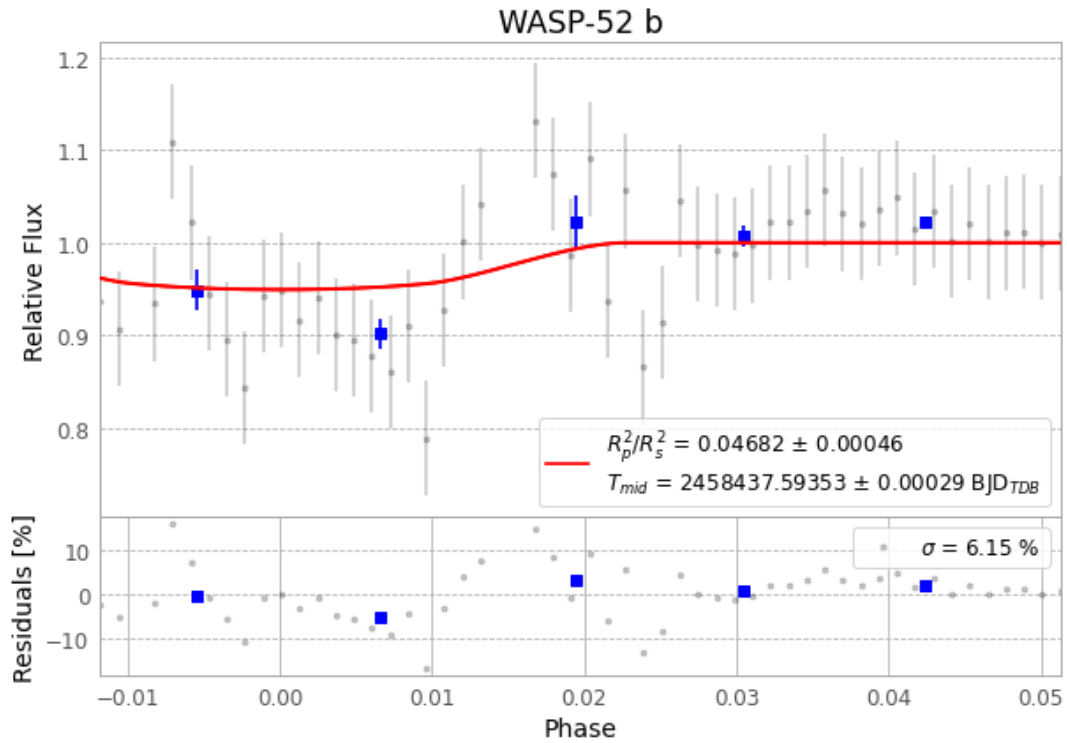


Figure 1 — FinalLightCurve_WASP-52 b_2018-11-14.png (SHA-256 6731cb655eb11a73...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c0c3a7e7209ec585...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

54 FITS frames (35 MB), WASP-52181115013912.FITS → WASP-52181115041812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-181115.log (d4978ffcbaac...), FinalLightCurve_WASP-52 b_2018-11-14.png (6731cb655eb1...), FinalLightCurve_WASP-52 b_2018-11-14.csv (9cd80eb573b1...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 07:03Z.